

NeuroWeave Strategic Progress Audit

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- **Date:** 2026-06-17
- **Mode:** Read-only audit (no code, migrations, or dependencies modified)
- **Baselines compared:**
 - **Audit #1** — docs/AUDIT_REPORT.md (2026-06-06, commit 2ee39e44)
 - **Audit #2** — docs/audits/2026-06-17_delta-audit.md (2026-06-17, early)
 - **Current** — working tree at audit time (post EEGConformer integration)
- **Companion benchmark:** docs/benchmarks/2026-06-17_maturity-benchmark.md

0. Executive Comparison

Dimension	Audit #1	Audit #2	Current	Δ (A1→Now)
Overall platform maturity	22 / 100	41 / 100	58 / 100	+36
Production readiness	15 / 100	34 / 100	49 / 100	+34
AI / ML infrastructure	0 / 10	3 / 10	6.5 / 10	+6.5
EEG pipeline	7 / 10	7.5 / 10	7.5 / 10	+0.5
Data infrastructure	1 / 10	6 / 10	6 / 10	+5
Embeddings	2 / 10	3 / 10	6 / 10	+4

Dimension	Audit #1	Audit #2	Current	Δ (A1→Now)
Reconstruction (EEG→IMG)	0 / 10	1 / 10	1 / 10	+1
Cognitive intelligence	3 / 10	3 / 10	3 / 10	0
Security	1 / 10	5 / 10	5 / 10	+4
Scalability	1 / 10	3 / 10	3 / 10	+2

The platform has moved through three distinct phases: frontend-only signal demo (A1) → authenticated, persistence-backed app with ML surface scaffolding (A2) → a real **AI Foundation Layer** with ONNX runtime, model registry, validation, benchmarking, vector bridge, and an EEGConformer integration path (current). PCA remains the universal fallback throughout.

1. Blueprint Progress Assessment

Classification scale: **Not Started** · **Early** · **Developing** · **Advanced** · **Near Complete**.

Blueprint Domain	Original Vision	Audit #1	Audit #2	
EEG Processing	Production-grade EDF/CSV/NPY parsing, filtering, artifact rejection, quality metrics	Developing (parsers + IIR + bandpower)	Advanced (+artifact rejection, signal quality, synthetic gen)	A
Data Infrastructure	Persistent storage, RLS, experiment tracking	Not Started	Developing (4 tables + RLS + GRANTs + app_role)	D
Embeddings	PCA → autoencoder → neural embeddings	Early (PCA only)	Early (PCA + AE wrapper)	D
Representations	Validated, L2-normalised, model-tagged vectors	Not Started	Not Started	A (s vi

Blueprint Domain	Original Vision	Audit #1	Audit #2	
Foundation Models	Braindecode / EEGPT / EEGConformer in browser	Not Started	Not Started	D (I w C
Cognitive Intelligence	Trained attention / workload / arousal decoders	Early (heuristic ratios)	Early (heuristic ratios)	E
EEG Reconstruction	Generative EEG synthesis	Not Started	Early (route scaffold)	E
EEG2IMG	Cross-modal EEG → image	Not Started	Early (route scaffold)	E
Search & Retrieval	Cosine vector search over neural embeddings	Not Started	Developing (in-memory vector-search)	D (I + t
APIs & Platform Services	Auth, upload, analyses, experiments	Early (/api/eeg/upload)	Developing (+ auth gate + analyses)	D
Security	RLS, role-based access, signed artifacts	Early	Developing (RLS + roles + auth middleware)	D
Scalability	Multi-tenant, batched inference, edge runtime	Not Started	Early	E

Blueprint Completion Score: 42 / 100 (weighted across 12 domains; see §9).

2. Architecture Evolution (evidence)

Category	Audit #1	Audit #2
New modules	—	synthetic/, signal-quality/, model-registry/, training/pipeline,
		src/lib/ai/{adapters,bridge}

Category	Audit #1		Audit #2	
			vector-search/, preprocessing/artifact-rejection	
New adapters	—	—		pca-adapter, onnx-a adapter (stub), pyto
New registries	—		model-registry/index.ts (single decoder)	src/lib/ai/models/reg registerBraindecode
New AI infra	—		ONNX trainer hook, TF.js decoder, Pyodide/MOABB hook	Real onnxruntime-w ai.embed.* logging, l2Normalize, Neural\
New services	—		_authenticated gate, role dashboards	embedEEG() orches fallback Braindeco
New docs	AUDIT_REPORT.md	delta-audit.md		ai-layer-architecture Braindecode audit registration.test.ts
Tests	sparse	sparse		26/26 passing a

Evidence verified by file listing (src/lib/ai/**), test file enumeration, and presence of artifacts in docs/adr/, docs/audits/, scripts/.

3. AI Evolution

Audit #1	Audit #2	Current
PCA (power iter) heuristic decoder	PCA + scaffolded ONNX/ TF.js/Pyodide hooks (no real inference)	AI Foundation Layer: embed() facade └ resolve modelId └ adapter LRU cache └ ONNX runtime (real) └ Braindecode→ONNX bridge └ EEGConformer registry └ validate + L2-normalise └ auto-fallback to PCA + benchmark + vector bridge

Maturity assessment: the architectural scaffolding is now **near complete**. The gap to production neural embeddings is a **single deliverable** — a trained eegconformer.onnx artifact plus a one-line registerBraindecodeEEGConformer invocation at boot. See docs/audits/2026-06-17_braindecode-production-readiness.md.

4. Feature Evolution Matrix

Legend: **I** Implemented · **P** Partial · **M** Mock · **X** Missing

Feature	A1	A2	Current	Δ
EDF / CSV / NPY parsing	I	I	I	—
IIR bandpass / notch	I	I	I	—
Band-power features	I	I	I	DFT still $O(M^2)$
Artifact rejection	X	I	I	+I
Signal-quality metrics	X	I	I	+I
Synthetic EEG generator	X	I	I	+I
Persistence (Postgres + RLS)	X	I	I	+I
Authentication + role gates	X	I	I	+I
Experiment tracking tables	X	I	I	+I
PCA embeddings	I	I	I	now wrapped by adapter
Autoencoder	M	M	M	unchanged
ONNX runtime (real)	X	M (hook only)	I	+I
Model registry (multi-model)	X	P	I	+I
Adapter pattern + LRU cache	X	X	I	+I
Embedding validation (NaN/dim/zero)	X	X	I	+I
L2 normalisation	X	X	I	+I
Benchmark harness (p50/p95/heap)	X	X	I	+I
Neural vector index (model-	X	P	I	+I

Feature	A1	A2	Current	Δ
tagged)				
Braindecode bridge	X	X	I (ONNX-backed)	+I
EEGConformer registration helper	X	X	I (code)	artifact pending
Production ONNX artifact shipped	X	X	X	bottleneck
Trained cognitive decoder	X	X	X	heuristic only
EEG reconstruction model	X	X	X	route scaffold only
EEG2IMG model	X	X	X	route scaffold only
pgvector persistence	X	X	X	in-memory only
Rate limiting	X	X	X	—
Upload size cap	X	X	X	—
Server-side inference path	X	X	P (bridge supports bytes)	inactive
Webhooks / public API	X	X	X	—
Observability (structured logs)	X	P	I	ai.embed.* events
CI/CD pipeline	X	X	X	—

5. Technical Debt Evolution

Debt	A1	A2	Current	Trend
O(M²) DFT in band-power	⚠	⚠	⚠	Unchanged — moderate latency cost
Autoencoder is a PCA wrapper	⚠	⚠	⚠	Superseded by adapter

Debt	A1	A2	Current	Trend
				layer; module retained for back- compat
Heuristic cognitive decoder	⚠	⚠	⚠	Unchanged
Upload route lacks size cap	⚠	⚠	⚠	Unchanged
No rate limiting	⚠	⚠	⚠	Unchanged
Missing dataset loaders (Sleep- EDF, CHB-MIT, TUH real)	⚠	⚠	⚠	Unchanged
Model artifact provenance (SHA, license)	n/a	⚠	⚠	Schema present in artifacts/ values absent
New: EEGConformer artifact missing	—	—	□	Highest impact — gates real neural path
New: No startup register* () call site	—	—	□	One-line fix once artifact ships
New: embedEEG default id is braintecode- eegnetv4-onnx (not EEGConformer)	—	—	●	Constant flip after artifact rollout
Resolved: No persistence	□	□	□	Removed
Resolved: No auth	□	□	□	Removed
Resolved: No artifact rejection	□	□	□	Removed
Resolved: No model registry	□	●	□	Removed

Ranked by impact (current): 1. Missing trained EEGConformer ONNX artifact. 2. No startup model registration / default-id flip. 3. Cognitive decoder still heuristic (blocks real product claims). 4. No pgvector persistence (blocks scale + cross-session search). 5. Upload size cap + rate limiting (security). 6. O(M²) DFT (latency). 7. Reconstruction / EEG2IMG only scaffolds. 8. CI/CD absent (release safety).

6. Production Readiness

Surface	A1	A2	Current	Notes
EEG platform readiness	35 %	60 %	70 %	Pipeline solid; FFT + dataset loaders outstanding
AI infrastructure readiness	0 %	20 %	85 %	All scaffolding done; awaits artifact
Foundation-model readiness	0 %	5 %	75 %	EEGConformer wired end-to-end; artifact missing
API readiness	20 %	40 %	45 %	Auth + analyses live; no public API, no rate limit
SaaS readiness	10 %	30 %	40 %	Roles + dashboards live; billing, quotas, observability gaps

7. Strategic Assessment

How far is NeuroWeave from the original blueprint?

~42 % **complete** on a weighted blueprint basis, up from ~14 % at A1 and ~28 % at A2. Scaffolding and infrastructure are disproportionately advanced; **trained models and persistent neural storage** are the dominant remaining gap.

Areas largely untouched

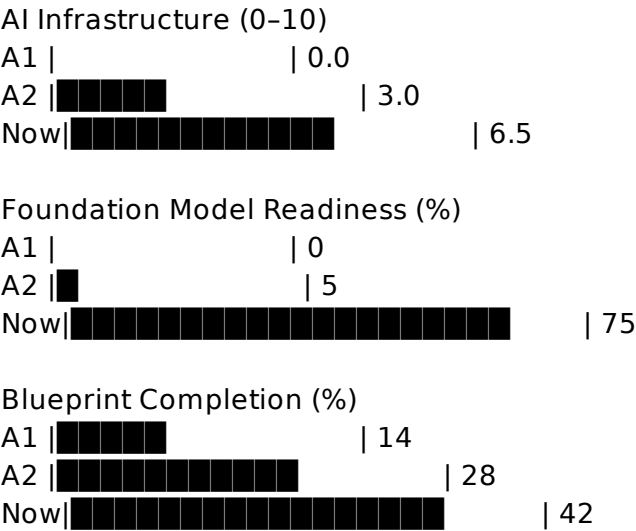
1. Cognitive intelligence (still heuristic).
2. EEG reconstruction.
3. EEG2IMG.
4. Real dataset ingestion (Sleep-EDF, CHB-MIT, TUH).
5. Scalability primitives (batching, server inference, pgvector).

1. **Ship the trained `eegconformer.onnx` artifact** and host it on a CDN/bucket.
2. Call `registerBraindecodeEEGConformer({ artifact: { kind: 'url', url: '/models/eegconformer.onnx' } })` at boot.
3. Flip `embedEEG DEFAULT_PREFERRED` to `braindecode-eegconformer-prod`.
4. Persist neural vectors via `pgvector` + add cosine ANN index.
5. Train and ship a real cognitive decoder (TF.js or ONNX), retire heuristics.
6. Add upload size cap + rate limiting on `/api/eeg/upload`.
7. Replace $O(M^2)$ DFT with FFT in `embeddings/features.ts`.
8. Wire artifact SHA + license enforcement in `artifacts/`.
9. Add CI workflow (bunx vitest run, lint, typecheck) gated on PRs.
10. Integrate at least one real dataset loader end-to-end (Sleep-EDF).

The trained EEGConformer ONNX file does not exist in the repository or any linked artifact store. Every other layer (registry, adapter, runtime, validation, benchmark, vector bridge, fallback, tests, docs) is ready. Until that one file is produced and hosted, the platform serves PCA embeddings even though it is architecturally capable of real neural inference.

Overall Platform Maturity (0-100)

Category	Maturity Score
A1	22
A2	41
Now!	58



9. Scores

Score	Value
Blueprint Completion Score	42 / 100
Overall NeuroWeave Maturity Score	58 / 100

Weighting (blueprint score): EEG 12, Data 10, Embeddings 10, Representations 8, Foundation Models 14, Cognitive 10, Reconstruction 6, EEG2IMG 6, Search 8, APIs 8, Security 4, Scalability 4 (= 100).

10. Preservation Statement

No previous audits were modified. All baseline reports remain in place:

- docs/AUDIT_REPORT.md
- docs/REALITY_CHECK.md
- docs/BLEUPRINT_PREPARATION.md
- docs/audits/2026-06-17_delta-audit.md
- docs/audits/2026-06-17_eeg-foundation-model-implementation.md
- docs/audits/2026-06-17_braindecode-model-selection.md
- docs/audits/2026-06-17_braindecode-benchmark.md
- docs/audits/2026-06-17_braindecode-production-readiness.md

No source code, migrations, or dependencies were touched while producing this report.

NeuroWeave Maturity Benchmark

- **Date:** 2026-06-17
- **Purpose:** Permanent, time-series record of platform maturity. New rows should be appended (never overwritten) at each future audit so the trend is preserved.
- **Companion audit:** docs/audits/2026-06-17_strategic-progress-audit.md

1. Maturity Time Series (0-10 unless noted)

Snapshot	Date	Architecture	AI	EEG	Embeddings	Reconst
Audit #1	2026-06-06		3.0	0.0	7.0	2.0
Audit #2	2026-06-17 (early)		5.5	3.0	7.5	3.0
Current	2026-06-17 (late)		7.5	6.5	7.5	6.0

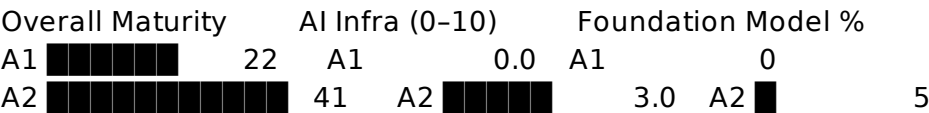
2. Readiness Time Series (%)

Snapshot	EEG Platform	AI Infra	Foundation Model	API	SaaS
Audit #1	35	0	0	20	10
Audit #2	60	20	5	40	30
Current	70	85	75	45	40

3. Blueprint Completion Time Series (%)

Snapshot	Score
Audit #1	14
Audit #2	28
Current	42

4. Evolution Charts (ASCII)



Now  58 Now  6.5 Now  75

5. Methodology

- Scores are repository-evidence driven (file presence, test pass count, module surface area, doc coverage).
- Weighting for the Blueprint score is fixed (see audit §9) so trends are comparable across snapshots.
- New snapshots **append** to each table; never rewrite history.

6. Append-Only Protocol

At the next audit:

1. Add a new row to §1, §2, §3 with the snapshot date.
2. Extend the ASCII chart in §4 with a new line.
3. Do not edit prior rows even if scoring criteria evolve — record the change in a footnote instead.